

Data Collection and Preprocessing Phase

Date	15 August 2026
Project Title	AI Career Pulse: Decoding the Job Market

Data Exploration and Preprocessing

Section	Work Completed / Findings
Data Overview	The dataset consists of 1,500 individual AI job postings described across 25 attributes, spanning postings from 2025 and 2026. It captures job titles across 12 job categories, 4 experience bands, 5 education levels, salary figures, 20 cities across 14 countries, remote-work arrangements, 5 company-size bands, 12 industries, required-skill sets, demand and year-over-year growth scores, benefits ratings, and indicator flags for seniority, remote-friendliness, and LLM-focused roles.
Data Cleaning	A full check across all 25 fields confirmed there are no missing values, no duplicate rows, and no repeated job_id entries, so the raw dataset can be used as-is without imputation or de-duplication.
Data Transformation	The existing categorical and numeric fields are being kept in their original form so Tableau can work with them directly; any additional calculated fields, such as salary bands or growth indices, will be derived inside Tableau as specific visualizations require them.
Data Type Conversion	Each column's data type was checked against its content: monetary and score-based fields (salary, demand, growth, benefits, years of experience) are stored as numeric values, while descriptive attributes (job title, category, location, education) remain as text. The posting_year and posting_month fields are numeric, allowing postings to be grouped and trended by month and year.
Column Splitting and Merging	The source file already separates job title, job category, city, country, experience level, salary components (minimum, maximum, and annual), industry, and other key dimensions into their own columns, so no further splitting or merging of fields was necessary.
Data Modeling	Because all 25 attributes live in a single flat table with one row per job posting, no joins or relational modelling are required; the existing columns can be mapped directly to Tableau dimensions, measures, filters, and calculated fields.
Save Processed Data	A cleaned and validated copy of the dataset has been stored ahead of the Tableau analysis stage. Data Set: https://drive.google.com/file/d/18rWhaGZbwtkOa65T4ti7aUuof2LQQZvg/view?usp=drivesdk